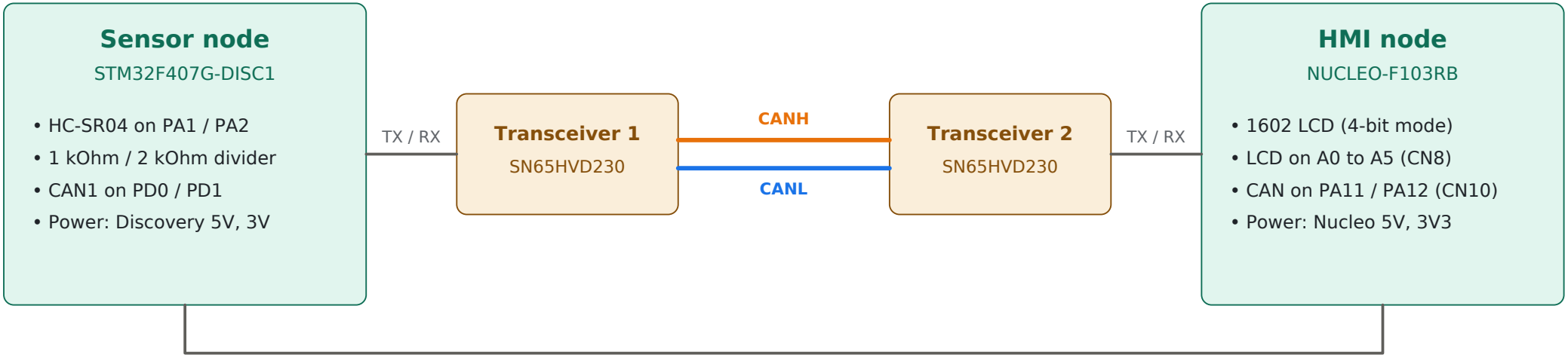
CAN Parking Assist: Wiring Reference

Sensor node (STM32F407G-DISC1) + HMI node (NUCLEO-F103RB) on one breadboard, connected over CAN at 500 kbit/s



Common GND wire between the boards (only GND, never 5V)

Rules that prevent most faults

1. Wire with both boards unplugged. Plug in one board at a time and check it before adding the next.
2. Never connect the 5V of the two boards together. Each board is powered from its own USB cable.
3. Transceivers run on 3.3 V (3V / 3V3 pin), never 5 V.
4. CAN TX goes to CAN TX and RX to RX. A CAN transceiver does not cross them like UART.
5. Every part sits with each pin in its own numbered breadboard column. A part turned 90 degrees shorts its pins.
6. Bus rails on large breadboards can be split in the middle. Bridge both halves or keep a node on one half.
7. The HC-SR04 echo is 5 V: it must reach PA1 only through the 1 kOhm / 2 kOhm divider.

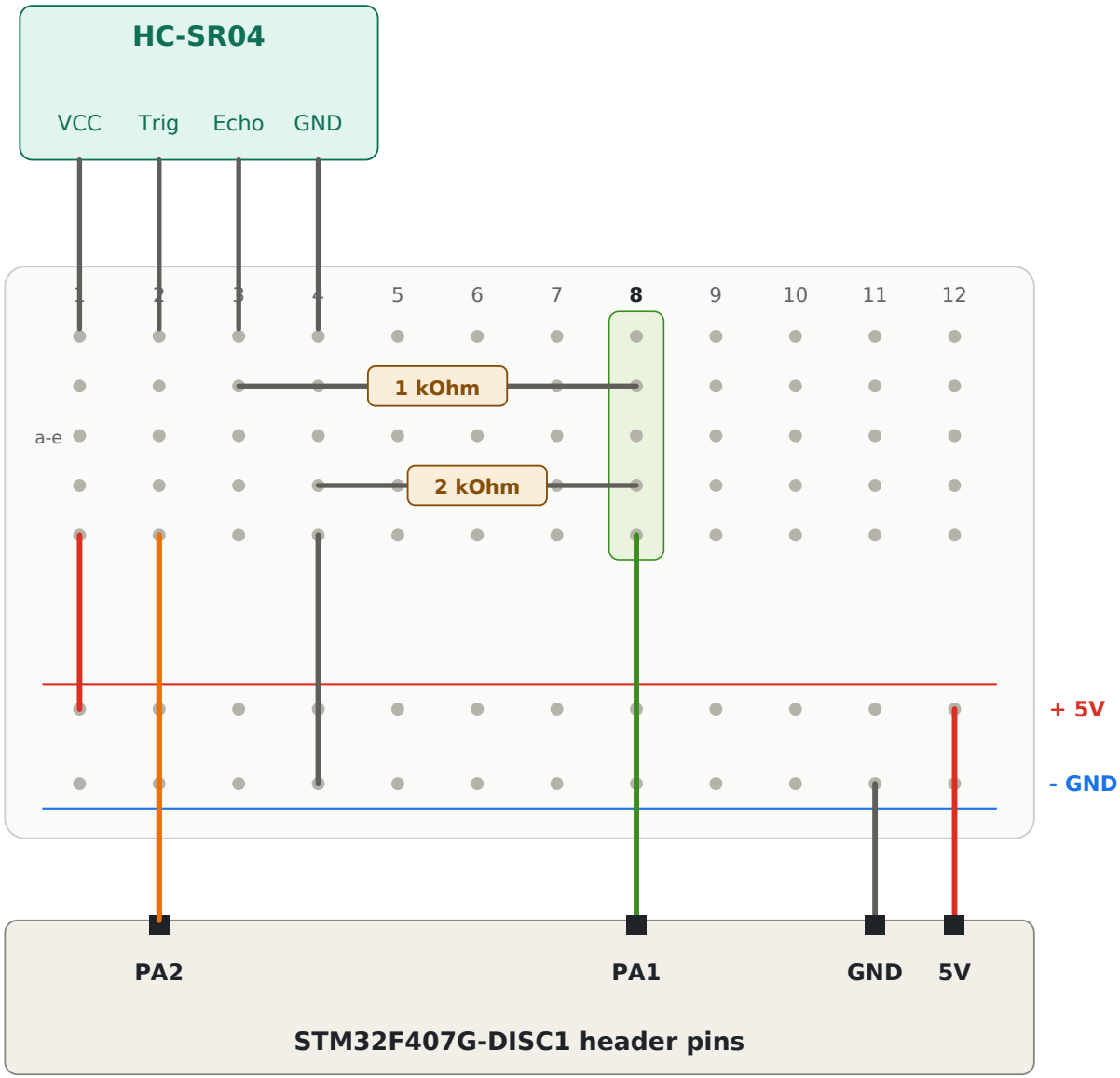
Suggested breadboard layout (large Wanjie board)

Area	Use
Bus strip A (top)	Discovery 5V (+) and GND (-)
Terminal strip 1	HC-SR04, divider, transceiver 1
Bus strip B	leave empty (separates the two supplies)
Terminal strip 2	1602 LCD, transceiver 2
Bus strip C	Nucleo 5V (+) and GND (-)
Strips 3 / 4, D / E	free (buzzer, LEDs later)

In every bus strip: upper row = +5V, lower row = GND. Mark the ends with tape.

1 Sensor node: HC-SR04 ultrasonic sensor

STM32F407G-DISC1 | Trig on PA2 (TIM2_CH3) | Echo on PA1 (TIM2_CH2) through a 5 V to 3.3 V divider



Column 8 (green) holds exactly three things: the 1 kOhm leg, the 2 kOhm leg and the PA1 wire.

Connections

From	To	Note
Sensor VCC	+ 5V rail	rail fed by Discovery 5V
Sensor GND	- GND rail	or direct to a GND pin
Sensor Trig	PA2	header P1, upper area
Sensor Echo	1 kOhm to column 8	never direct to PA1
Column 8	PA1	divider midpoint, 3.3 V
Column 8	2 kOhm to sensor GND	lower divider half
Discovery 5V	+ 5V rail	top of header P2
Discovery GND	- GND rail	any GND pin

Voltage divider

```
Echo 5V --[1k]--+-- PA1 (about 3.3 V)
                |
              [2k]
                |
               GND
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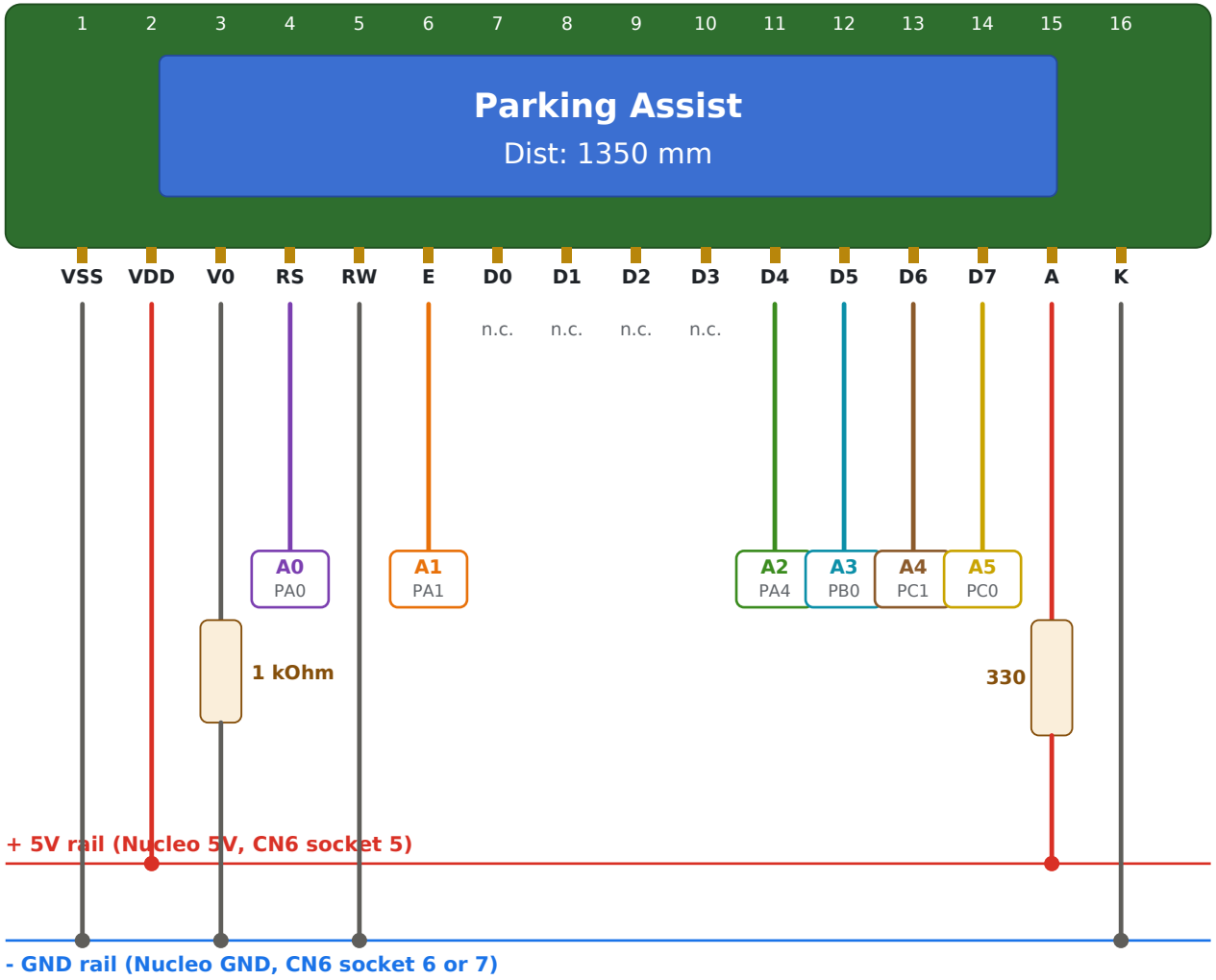
Resistor colour codes (5-band, blue body)

Value	Bands	Quick check
1 kOhm	brown black black brown brown	no red band
2 kOhm	red black black brown brown	red first band
330 Ohm	orange orange black black brown	LCD backlight

Software: TIM2 prescaler 83 (1 us/tick), period 59999 (60 ms),
CH3 PWM pulse 10 (10 us trigger), CH2 input capture both edges.
Distance mm = echo us x 343 / 2000.

2 HMI node: 1602 LCD (HD44780, 4-bit mode)

NUCLEO-F103RB | data on A0 to A5 (header CN8) | 5 V supply from CN6 | no potentiometer needed



Boxes = wire to that Nucleo pin (Arduino label / MCU pin). n.c. = not connected.
Each of the 16 LCD pins must sit in its own breadboard column.

Connections

Pin	Name	Connect to
1	VSS	- GND rail
2	VDD	+ 5V rail
3	V0	1 kOhm to GND (contrast)
4	RS	A0 (PA0), CN8
5	RW	GND (write only, protects MCU)
6	E	A1 (PA1), CN8
7-10	D0-D3	not connected
11	D4	A2 (PA4), CN8
12	D5	A3 (PB0), CN8
13	D6	A4 (PC1), CN8
14	D7	A5 (PC0), CN8
15	A	330 Ohm to + 5V rail
16	K	- GND rail

Contrast without a potentiometer

- 1 kOhm V0 to GND: normal contrast (recommended)
- Blocks too faint: V0 directly to GND
- Too dark: use 2 kOhm instead
- V0 unconnected: display looks blank. Avoid.

Power-up check

Backlight on + a row of dark blocks on line 1
= power, contrast and wiring are correct.

3 CAN transceivers and bus, and how to bring it up

One SN65HVD230 module per node | 500 kbit/s | 120 Ohm termination on each module (60 Ohm across the bus)

Transceiver modules

Module pin	Sensor node (STM32F407)	HMI node (STM32F103)
3.3V	3V pin (P2, top)	3V3 (CN6, socket 4)
GND	GND	GND
CAN TX	PD1 (P2, lower half)	PA12: CN10 outer row, level with SCK/D13
CAN RX	PD0 (P2, lower half)	PA11: CN10 outer row, level with MISO/D12

Bus between the nodes

From	To	Note
Module 1 CANH	Module 2 CANH	twist with the CANL wire
Module 1 CANL	Module 2 CANL	2-pin header next to terminal
Discovery GND	Nucleo GND	common ground; never connect the 5V pins

Logic analyser (only when measuring)

Analyser	Connect to	Shows
GND	Discovery GND	reference, always needed
CH1	PD0	whole bus traffic incl. ACK bit
CH2	PD1	sensor node's own frames

Bring-up in stages

1 Sensor node only

Plug in the Discovery. Console shows distances, st=0.

2 Add the HMI node

Plug in the Nucleo. LCD shows 'Parking Assist'.

3 Bus

Discovery green LED on, Nucleo LED blinks, live distance.

If the bus does not work: read the Discovery LEDs

LEDs	Meaning	Check
Green	bus OK	nothing to do
Red + blue	no acknowledgement	HMI side: PA11/12, 3V3, bus
Red only	sensor cannot see bus	own module: PD0/PD1, 3V